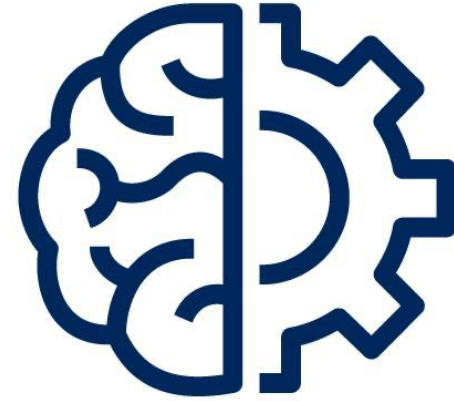
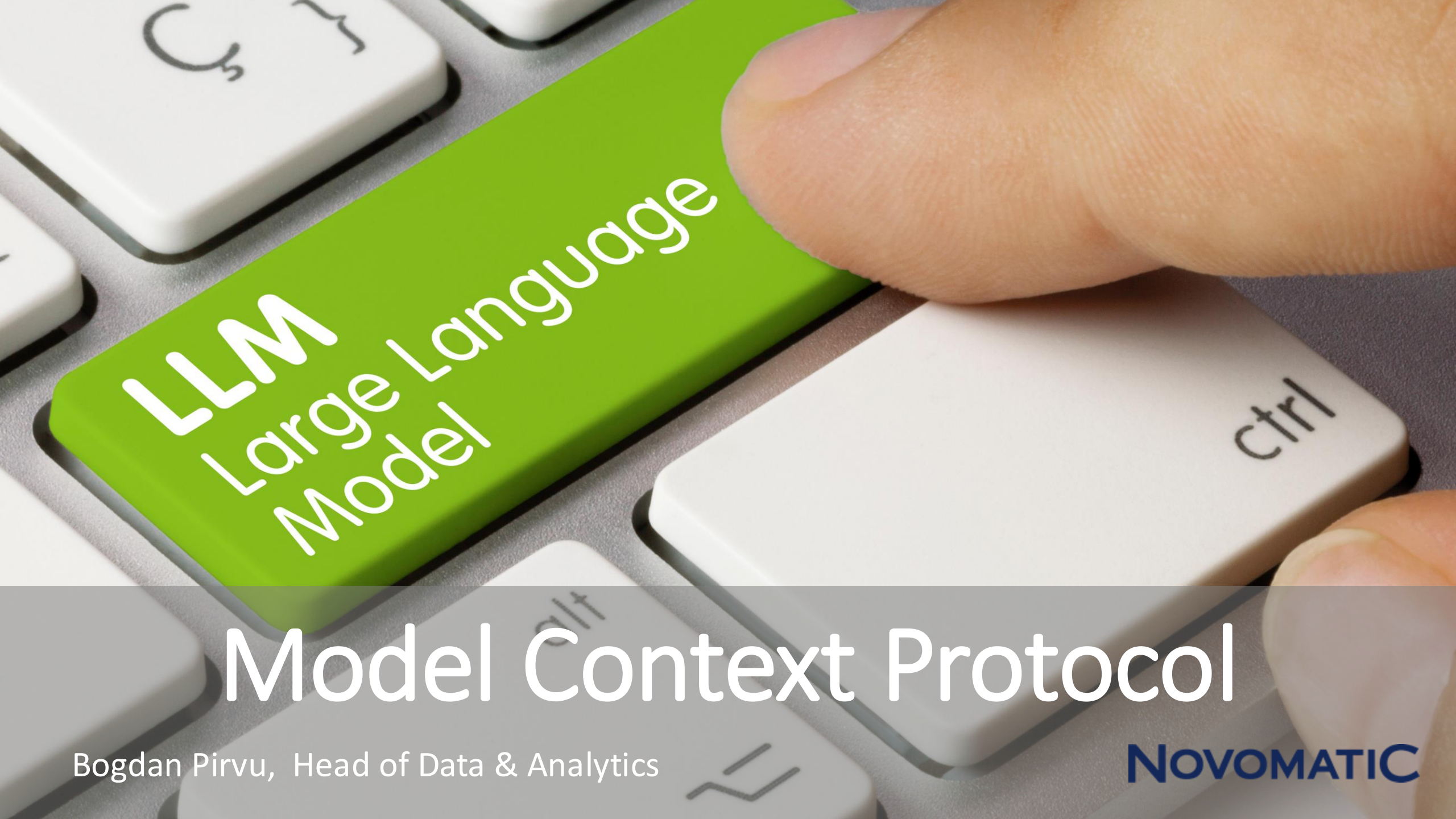


VIENNA AI ENGINEERING



We talk about the art and
craftsmanship of creating reliable
AI-powered applications
that users love



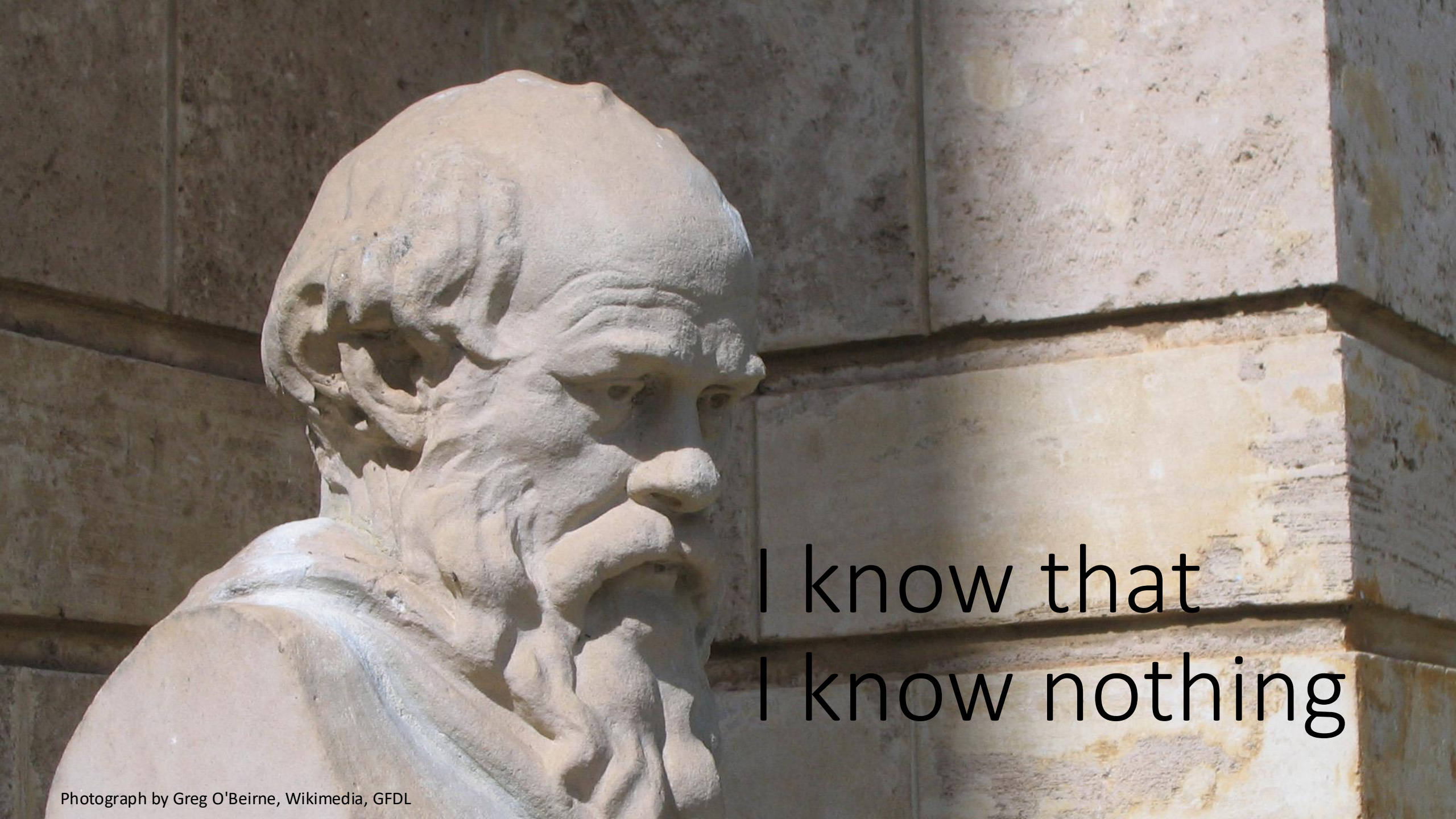


LLM
Large Language
Model

Model Context Protocol

Bogdan Pirvu, Head of Data & Analytics

NOVOMATIC



I know that
I know nothing

MCP

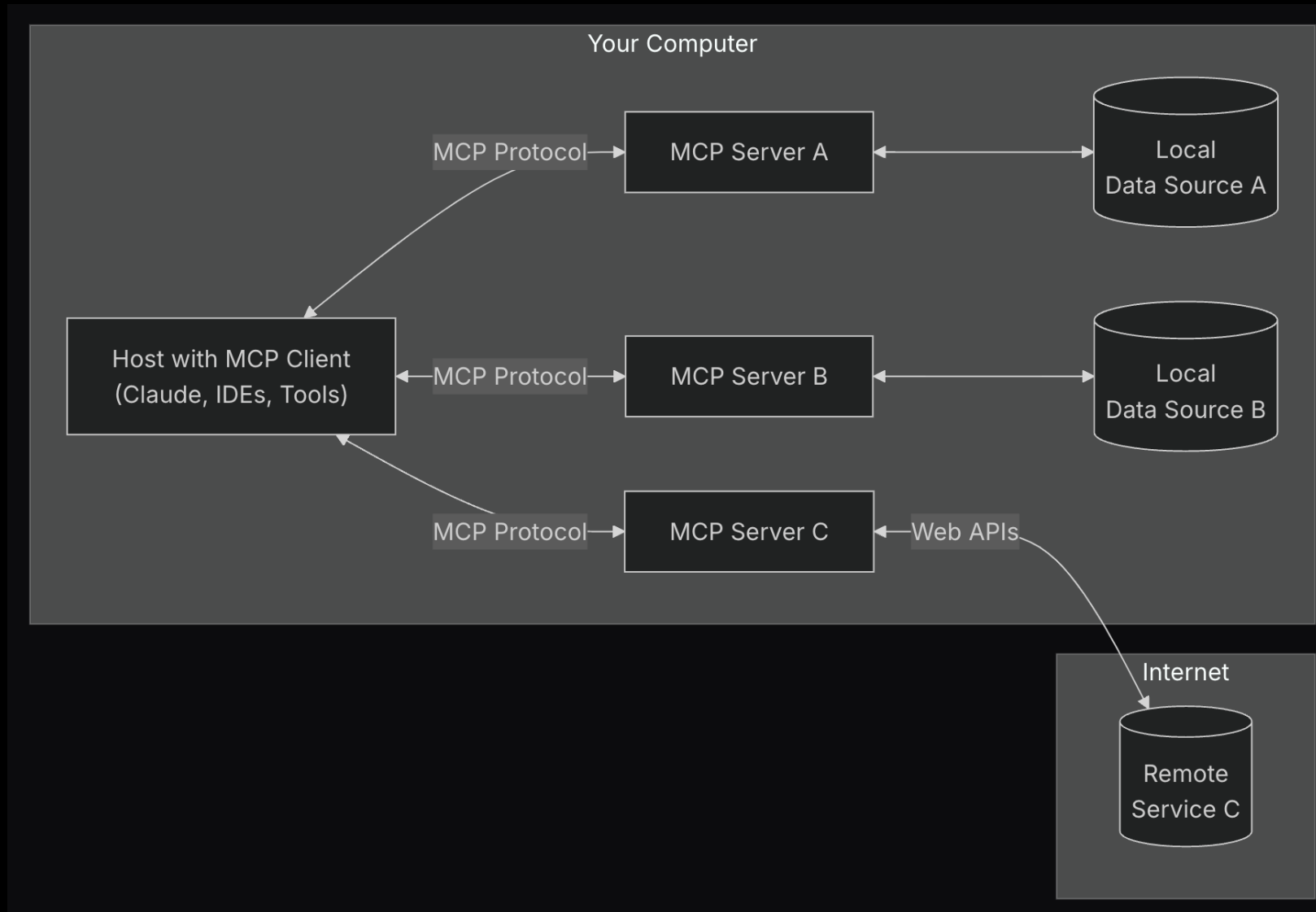
The Model Context Protocol

What is MCP?

- It's a protocol for seamless integration between LLMs and external context sources
- Structured context representation enabling dynamic and precise prompt augmentation
- Extensible schema supporting diverse data types and custom tool integrations
- Cross-platform compatibility ensuring consistent context handling across model providers
- Built-in governance features for context validation, provenance tracking, and secure data handling

A picture please...

The Model Context Protocol



1. Official Docs - "Introduction - Get started with the Model Context Protocol (MCP)", <https://modelcontextprotocol.io/introduction>

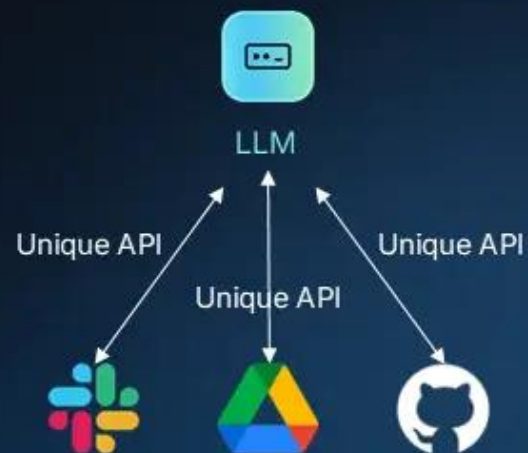
MCP Concepts

- Client-Server Architecture
 - Host (e.g. Claude Desktop)
 - Typically runs many clients
 - Server
 - Runs locally or remotely
- Transport layer
 - STDIO
 - HTTP with SSE
- Core Capabilities
 - Resources
 - Tools
 - Prompts

More pictures please...



Before MCP

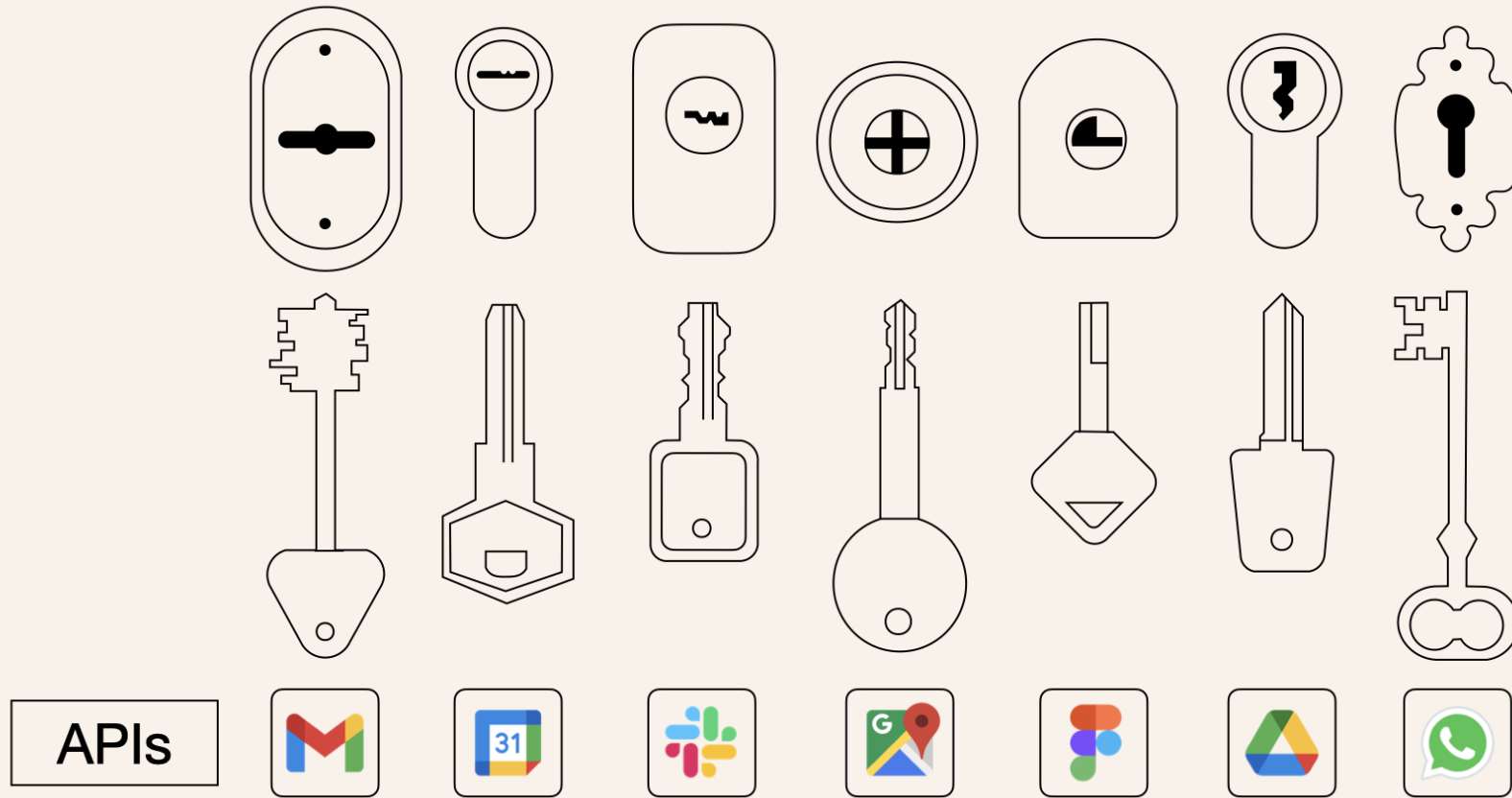


After MCP

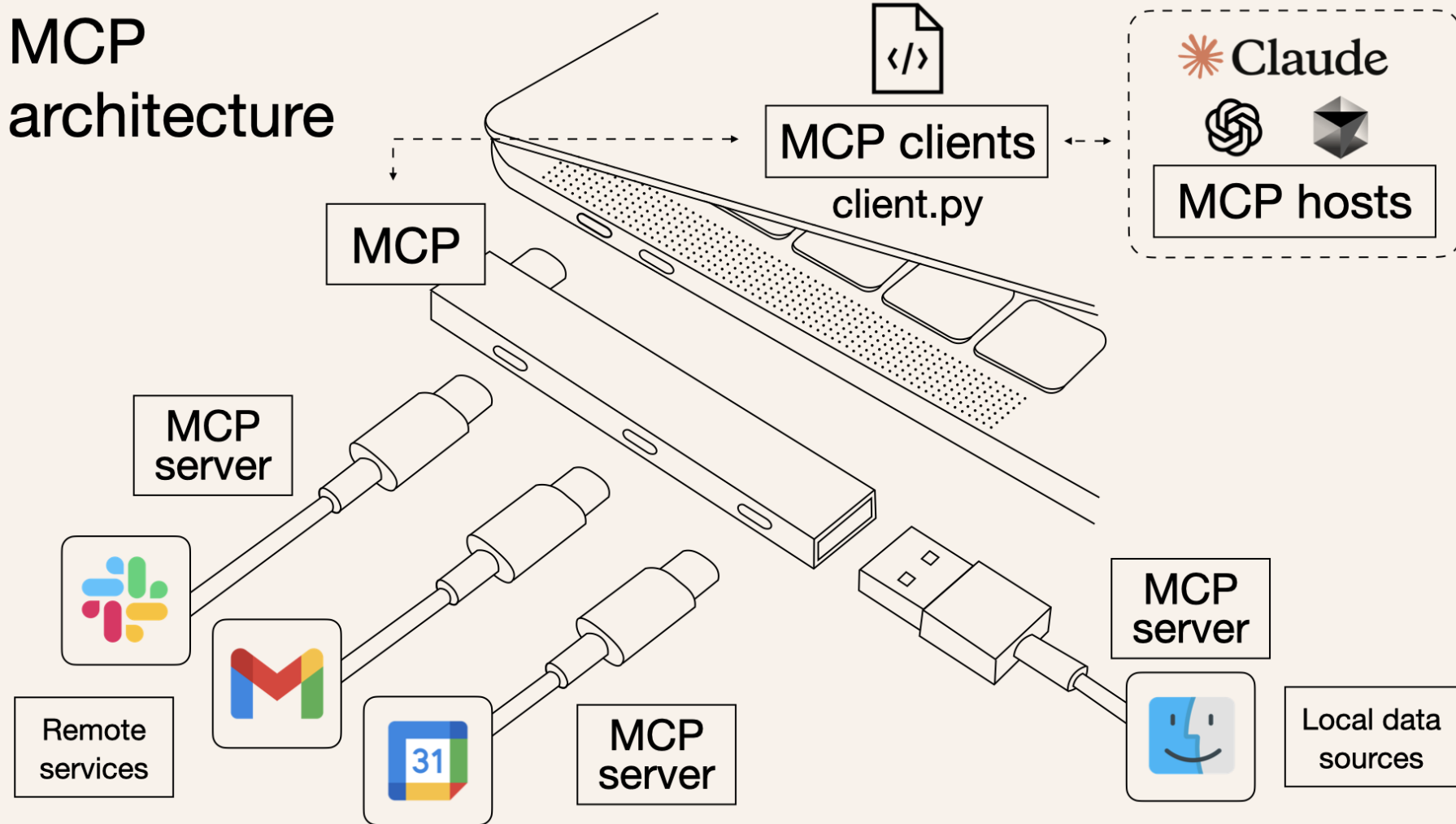


APIs: Every tool needs its own key

Traditional APIs require different authentication and integration for each service,
like needing different keys for different locks



MCP architecture



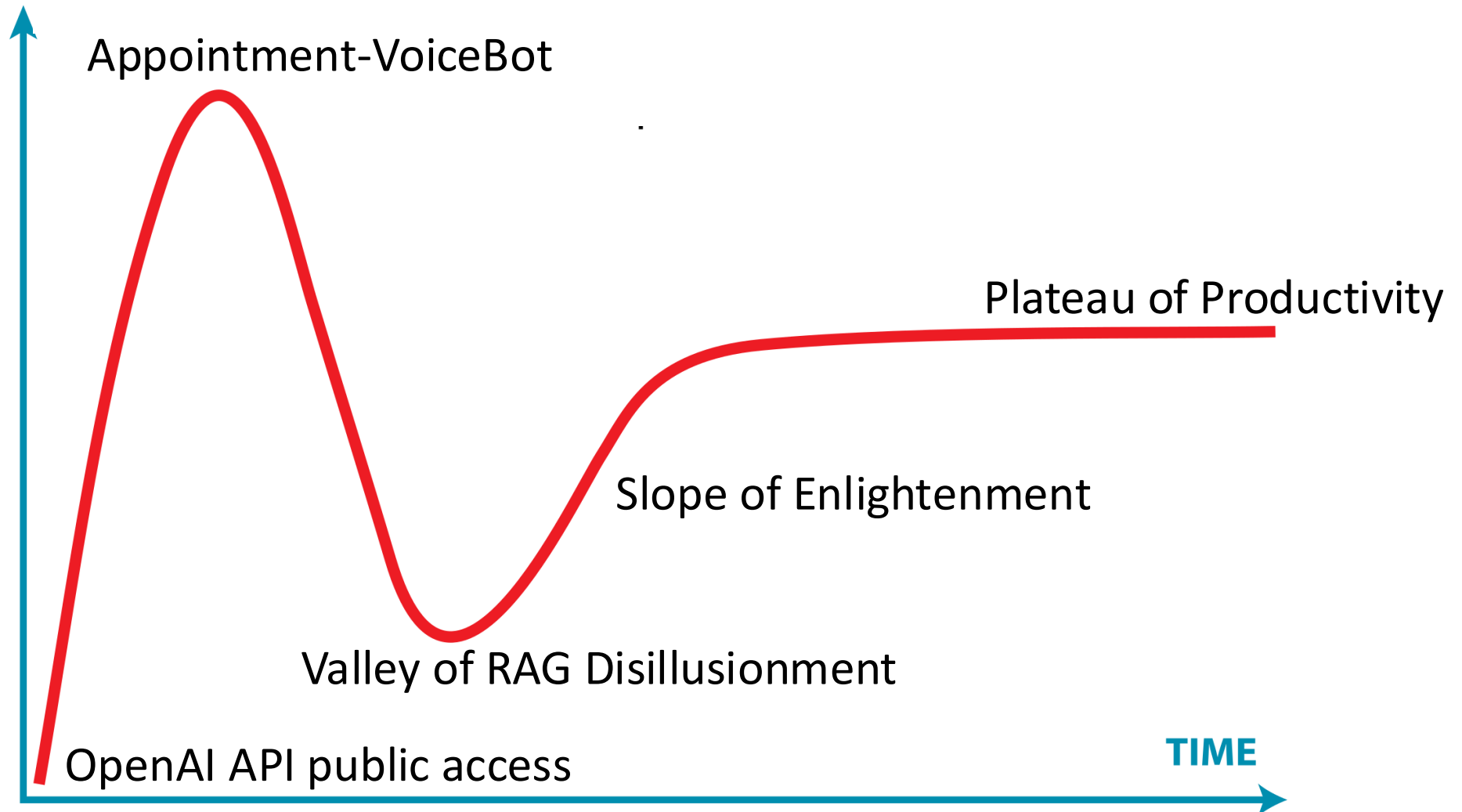
1. Blogpost - "What is (MCP)? How it simplifies AI integrations compared to APIs", <https://norahsakal.com/blog/mcp-vs-api-model-context-protocol-explained/>

Isn't that great?

What could possibly stop us now?



The GenAI Reality



Demo

Appointment VoiceBot with OpenAI, Twilio, Google Calendar

Show Google Calendar

Demo

Appointment Bot with an OpenAI GPT

[Show Google Calendar](#)

Demo

Appointment Bot with MCP in Claude Desktop

Show Google Calendar

Demo

Arithmetic Bot with MCP in Claude Desktop

[Show Claude Desktop](#)

Main Problems

- MCP is not very mature yet
- Breaking changes¹
- Messy implementation, even in Claude Desktop²
- Not all channels supported in client apps, even in Claude Desktop³
 - Local Proxy:
 - <https://www.npmjs.com/package/mcp-remote>

1. MCP Java SDK Docs, <https://modelcontextprotocol.io/sdk/java/mcp-overview>
2. Locally started MCP Servers persist as orphaned processes after closing Claude Desktop
3. HTTP-with-SSE not directly supported, only available via local proxy app

Conclusion

- Great idea
- Great potential
- Needs to grow up
- And I'm convinced it will, eventually...



LLM
Large Language
Model

Thank you!

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Stay updated on LLMs and more



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